

Jason Quemard

jasonq04@hotmail.co.uk - (+44) 7394 39 3321 - Based in Hampshire

Profile

First class BEng Motorsport/Mechanical Engineering graduate with practical experience in CAD, reverse engineering and mechanical design. Enjoys conceiving, developing and prototyping complete mechanical solutions. Seeking a graduate role in design engineering and/or fabrication.

Technical Experience

Engineering Design for Manufacture, Prototyping, Drawings & CAD/CAE – High level of competency in CAD across SolidWorks, Fusion 360 and Blender. Modelling automotive chassis, bodies, system package layouts, pumps, suspension systems, wheels and bespoke mechanisms. Able to read and produce engineering drawings.

- ❖ *SSGT project*: Designed the concept for a Single Seater Grand Touring vehicle, fully defining the package layout and drivetrain architecture. Performed a full kinematic and dynamic suspension design and optimisation using MSC ADAMS for BEng dissertation.
- ❖ *EPAC project*: Designed a novel compact crank system for a recumbent EPAC, halving the frontal area, improving the aerodynamic and space efficiency. Applied DFM principles for sheet metal fabrication, manual lathe turning and end milling, and utilised standardised, off-the-shelf components.
- ❖ *RC project*: Designed, modelled and manufactured a complete 1:10 scale car chassis, gearbox and suspension system. Iteratively tested and improved the chassis, suspension and aero package.

Reverse Engineering – Rapidly recreated parts and assemblies in CAD for component reproduction and adapting new functionality.

- ❖ *Tool Reproduction*: Reverse-engineered and produced an obsolete tool required by a client, reducing cost of acquisition by 99% and cutting delivery time from 1 week to 1 hour.
- ❖ *Gearbox Redesign Project*: Reverse-engineered and reproduced a commercial Traxxas gearbox assembly to optimise the mounting points for chassis integration and greatly improve repair time and expense.
- ❖ *Riley Kestrel 1100 Mesh Insert Project*: Reverse-engineered the cowl vent of an 1100 to design and produce a novel mesh insert that eliminates a design flaw causing premature rusting, while integrating with the aesthetics of the vehicle, validated by the chairman of the 1100 club.

Work Experience

Oxford Universities Motorsport Foundation (OUMF) – Member

OUMF is a volunteer student run foundation that works on restoring, modifying and racing classic rally and circuit cars.

- ❖ *Team Environment*: Operated as part of a team to coordinate on projects and brainstorm design ideas.
- ❖ *Independent Tasks*: Assisted with workshop organisation/cleanliness and event preparation.
- ❖ *Bracketry*: Designed and fabricated bespoke sheet metal and 3D printed bracketry used on the race cars.
- ❖ *Vehicle Maintenance*: Performed post-race checks on the vehicles, painted parts and assisted with engine rebuilding and vehicle maintenance/repair.

Royal Mail – Temporary Mail Sorter (12/2023 - 01/2024)

McDonalds – Team Member (01/2021 – 08/2022)

Education

Oxford Brookes University: BEng Motorsport Engineering, 1st Class Honours (08/2022 - 05/2026)

- ❖ *Machine Shop Experience*: Undertook a class on creating a Stirling engine from drawings involving deburring CNC components, manual lathe turning piston bores, drilling holes, hand tapping and assembly.
- ❖ *Composites Experience*: Achieved an A+ in materials engineering, covering design principles and manufacturing of composite structures. Performed hand lay-up and vacuum bagging of prepreg carbon fibre to create differently oriented samples for use in tensile testing.
- ❖ *MATLAB and Simulink*: Used MATLAB to create a steady-state vehicle dynamics lap-time simulator, and as part of dissertation to perform ride frequency calculations. Created PID control loops in Simulink.

Barton Peveril Sixth Form College (08/2020 – 05/2022), A-Levels: Maths, Physics, Computer Science

Wyvern Technology College (08/2016 – 05/2020), GCSE: English Language, English Literature, Maths and Science.

Additional Information

- ❖ Portfolio available @ www.poss47.github.io/i
- ❖ Languages: English (native), Turkish (fluent)